

Qualification Pack



Machine Operator - CNC Milling of Plastic

QP Code: RSC/Q8401

Version: 3.0

NSQF Level: 4

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RSC/Q8401: Machine Operator - CNC Milling of Plastic

Brief Job Description

The Machine Operator - CNC Milling of Plastic is responsible for interpreting CNC programs and engineering drawings, setting up CNC milling machines, performing milling operations on plastic workpieces, and ensuring precision and quality of the final products. The role involves selecting appropriate tools, adjusting machine parameters, conducting inspection checks, and performing routine maintenance. Operators must adhere to workplace safety standards, follow sustainable practices, and maintain the work area in optimal condition.

Personal Attributes

The job requires the individual to be calm, composed and physically fit along with good observational skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [RSC/N9901: Interpret CNC programs and drawings and set up CNC Milling Machine](#)
2. [RSC/N9902: Perform milling operations on plastic workpieces using CNC Machine](#)
3. [RSC/N9903: Maintain machine, tools, and work area](#)
4. [RSC/N5007: Follow health & safety protocol](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Rubber
Sub-Sector	Tool and Die Moulding
Occupation	CNC Milling
Country	India
NSQF Level	4
Credits	17

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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7223.5004
Minimum Educational Qualification & Experience	12th grade Pass (Or Equivalent) OR 10th grade pass plus 2-year NTC (/NAC/CITS or Equivalent) OR 10th grade pass with 3 Years of experience Relevant OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	30/04/2028
NSQC Approval Date	30/04/2025
Version	3.0
Reference code on NQR	QG-04-RI-04048-2025-V2-RSDC
NQR Version	2

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RSC/N9901: Interpret CNC programs and drawings and set up CNC Milling Machine

Description

This NOS covers the skills and knowledge required to read and interpret CNC programs and engineering drawings, select appropriate cutting tools and workholding devices, set up CNC milling machines for plastic machining, and ensure that machine parameters are configured according to the job specifications.

Scope

The scope covers the following :

- Interpret job requirements
- Prepare machine setup
- Configure machine parameters
- Conduct trial run and verification

Elements and Performance Criteria

Interpret job requirements

To be competent, the user/individual on the job must be able to:

- PC1.** obtain job order and specifications from the authorized person
- PC2.** read and interpret CNC programs for the specified plastic component
- PC3.** read engineering drawings and identify dimensions, tolerances, and surface finish requirements
- PC4.** identify material type and properties of the plastic workpiece to be machined.

Prepare machine setup

To be competent, the user/individual on the job must be able to:

- PC5.** select and verify the cutting tools as per job requirements
- PC6.** inspect workholding devices and fixtures for suitability
- PC7.** mount and secure workpiece on the machine table using approved clamping methods
- PC8.** install and align the selected cutting tools in the machine spindle
- PC9.** set workpiece datum and zero points according to the program reference

Configure machine parameters

To be competent, the user/individual on the job must be able to:

- PC10.** load and verify the CNC program in the machine controller
- PC11.** adjust spindle speed, feed rate, and depth of cut as per plastic machining requirements
- PC12.** simulate the CNC program and check for errors
- PC13.** ensure coolant or air blast is set appropriately for the plastic material

Conduct trial run and verification

To be competent, the user/individual on the job must be able to:

- PC14.** perform a dry run without cutting to verify tool paths

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- PC15.** carry out a trial cut and measure the output against specifications
- PC16.** make necessary parameter adjustments to achieve dimensional accuracy
- PC17.** ensure setup is approved before starting production

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic principles of CNC milling and coordinate systems
- KU2.** types of plastics and their machinability characteristics
- KU3.** interpretation of engineering drawings including symbols, tolerances, and surface finish
- KU4.** understanding of CNC program codes (G-codes, M-codes) relevant to milling
- KU5.** selection criteria for cutting tools for plastic machining
- KU6.** types and uses of workholding devices and fixtures
- KU7.** machine setup procedures for plastic milling
- KU8.** importance of correct datum setting and zero referencing
- KU9.** effect of spindle speed, feed rate, and depth of cut on plastic materials
- KU10.** common errors in CNC programs and methods to troubleshoot
- KU11.** safety measures during machine setup and trial runs
- KU12.** environmental considerations for plastic machining operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and interpret technical drawings and CNC programs
- GS2.** communicate effectively with supervisors and team members
- GS3.** use measurement tools such as calipers and micrometers
- GS4.** apply problem-solving skills to identify and correct setup issues
- GS5.** follow standard operating procedures and safety guidelines

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpret job requirements</i>	8	12	-	-
PC1. obtain job order and specifications from the authorized person	2	3	-	-
PC2. read and interpret CNC programs for the specified plastic component	2	3	-	-
PC3. read engineering drawings and identify dimensions, tolerances, and surface finish requirements	2	3	-	-
PC4. identify material type and properties of the plastic workpiece to be machined.	2	3	-	-
<i>Prepare machine setup</i>	12	20	-	-
PC5. select and verify the cutting tools as per job requirements	2	3	-	-
PC6. inspect workholding devices and fixtures for suitability	2	3	-	-
PC7. mount and secure workpiece on the machine table using approved clamping methods	2	4	-	-
PC8. install and align the selected cutting tools in the machine spindle	3	5	-	-
PC9. set workpiece datum and zero points according to the program reference	3	5	-	-
<i>Configure machine parameters</i>	12	16	-	-
PC10. load and verify the CNC program in the machine controller	3	4	-	-
PC11. adjust spindle speed, feed rate, and depth of cut as per plastic machining requirements	3	4	-	-
PC12. simulate the CNC program and check for errors	3	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. ensure coolant or air blast is set appropriately for the plastic material	3	4	-	-
<i>Conduct trial run and verification</i>	8	12	-	-
PC14. perform a dry run without cutting to verify tool paths	2	3	-	-
PC15. carry out a trial cut and measure the output against specifications	2	3	-	-
PC16. make necessary parameter adjustments to achieve dimensional accuracy	2	3	-	-
PC17. ensure setup is approved before starting production	2	3	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9901
NOS Name	Interpret CNC programs and drawings and set up CNC Milling Machine
Sector	Rubber
Sub-Sector	Plastic Post - Processing
Occupation	CNC Machining and Die Manufacturing
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N9902: Perform milling operations on plastic workpieces using CNC Machine

Description

This NOS covers executing CNC milling operations on plastic workpieces, monitoring the machining process, ensuring product quality, and making necessary adjustments to achieve dimensional accuracy and surface finish.

Scope

The scope covers the following :

- Prepare for machining operations
- Execute CNC milling operations
- Conduct in-process inspection and quality control
- Complete machining operations

Elements and Performance Criteria

Prepare for machining operations

To be competent, the user/individual on the job must be able to:

- PC1.** verify the CNC milling program against the work order and technical drawings
- PC2.** load the correct workpiece and secure it on the machine fixture or vice
- PC3.** check the availability and condition of required cutting tools and inserts
- PC4.** set tool offsets and work coordinates using precision measuring instruments
- PC5.** ensure coolant and lubrication systems are functioning as per the requirement

Execute CNC milling operations

To be competent, the user/individual on the job must be able to:

- PC6.** start the CNC program and monitor the initial machining pass
- PC7.** adjust feed rate, spindle speed, and depth of cut as per the material's behavior
- PC8.** monitor chip formation, tool wear, and surface finish during operations
- PC9.** apply emergency stop procedures in case of abnormalities
- PC10.** maintain consistent production cycle time while meeting quality parameters

Conduct in-process inspection and quality control

To be competent, the user/individual on the job must be able to:

- PC11.** measure dimensions using vernier calipers, micrometers, gauges, or CMM
- PC12.** compare measurements with specifications in the job order or drawing
- PC13.** make necessary offsets in the program to correct deviations
- PC14.** record in-process inspection data as per organisational procedures

Complete machining operations

To be competent, the user/individual on the job must be able to:

- PC15.** deburr and clean the machined workpiece

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- PC16.** unload finished parts and place them in designated storage areas
- PC17.** perform a visual inspection to detect any surface defects
- PC18.** hand over machined components with inspection report to the next stage

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** basic principles of CNC milling and its application in plastic processing
- KU2.** common types of plastics used in CNC milling and their machining properties
- KU3.** interpretation of CNC codes (G-code, M-code) and their functions
- KU4.** reading and interpreting technical drawings and tolerances
- KU5.** importance of tool selection, geometry, and cutting parameters for plastics
- KU6.** methods to set work and tool offsets accurately
- KU7.** significance of coolant types and lubrication in plastic machining
- KU8.** in-process quality inspection methods and acceptance criteria
- KU9.** common machining defects in plastics and their remedies
- KU10.** health, safety, and environmental precautions during CNC operations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate machining progress and issues effectively with supervisors
- GS2.** read and interpret machine manuals and program sheets
- GS3.** calculate dimensions, tolerances, and cutting parameters
- GS4.** use problem-solving skills to handle machining errors and deviations
- GS5.** work with precision and attention to detail

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for machining operations</i>	10	15	-	-
PC1. verify the CNC milling program against the work order and technical drawings	2	3	-	-
PC2. load the correct workpiece and secure it on the machine fixture or vice	2	3	-	-
PC3. check the availability and condition of required cutting tools and inserts	2	3	-	-
PC4. set tool offsets and work coordinates using precision measuring instruments	2	3	-	-
PC5. ensure coolant and lubrication systems are functioning as per the requirement	2	3	-	-
<i>Execute CNC milling operations</i>	14	20	-	-
PC6. start the CNC program and monitor the initial machining pass	3	4	-	-
PC7. adjust feed rate, spindle speed, and depth of cut as per the material's behavior	3	4	-	-
PC8. monitor chip formation, tool wear, and surface finish during operations	3	4	-	-
PC9. apply emergency stop procedures in case of abnormalities	3	4	-	-
PC10. maintain consistent production cycle time while meeting quality parameters	2	4	-	-
<i>Conduct in-process inspection and quality control</i>	8	13	-	-
PC11. measure dimensions using vernier calipers, micrometers, gauges, or CMM	2	4	-	-
PC12. compare measurements with specifications in the job order or drawing	2	3	-	-
PC13. make necessary offsets in the program to correct deviations	2	3	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. record in-process inspection data as per organisational procedures	2	3	-	-
<i>Complete machining operations</i>	8	12	-	-
PC15. deburr and clean the machined workpiece	2	3	-	-
PC16. unload finished parts and place them in designated storage areas	2	3	-	-
PC17. perform a visual inspection to detect any surface defects	2	3	-	-
PC18. hand over machined components with inspection report to the next stage	2	3	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9902
NOS Name	Perform milling operations on plastic workpieces using CNC Machine
Sector	Rubber
Sub-Sector	Plastic Post - Processing
Occupation	CNC Machining and Die Manufacturing
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N9903: Maintain machine, tools, and work area

Description

This NOS covers routine and preventive maintenance of CNC milling machines, upkeep of tools, and ensuring a clean, safe, and organized work area to enhance productivity and machine life.

Scope

The scope covers the following :

- Perform routine maintenance of CNC milling machine
- Maintain cutting tools and accessories
- Maintain cleanliness and safety of the work area

Elements and Performance Criteria

Perform routine maintenance of CNC milling machine

To be competent, the user/individual on the job must be able to:

- PC1.** clean the machine bed, fixtures, and tool holders after operations
- PC2.** check and top up coolant and lubrication levels as per maintenance schedule
- PC3.** inspect belts, filters, and hoses for wear and tear
- PC4.** verify spindle alignment and machine accuracy using test indicators
- PC5.** lubricate moving parts following manufacturer's recommendations

Maintain cutting tools and accessories

To be competent, the user/individual on the job must be able to:

- PC6.** inspect cutting tools for wear, chipping, or deformation
- PC7.** replace worn-out tools and update tool inventory records
- PC8.** store tools in a clean, dry, and organized manner to prevent damage
- PC9.** calibrate and store measuring instruments securely

Maintain cleanliness and safety of the work area

To be competent, the user/individual on the job must be able to:

- PC10.** remove chips, dust, and debris from the work area regularly
- PC11.** segregate and dispose of plastic waste and scrap as per guidelines
- PC12.** ensure all safety guards, covers, and emergency stops are functional
- PC13.** store hazardous materials in compliance with safety norms
- PC14.** maintain proper lighting and ventilation in the work area

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** preventive and routine maintenance procedures for CNC milling machines

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- KU2.** manufacturers' maintenance guidelines and recommended intervals
- KU3.** types of cutting tool wear and their impact on machining quality
- KU4.** correct storage methods for tools and measuring devices
- KU5.** safe disposal and recycling methods for plastic waste
- KU6.** importance of maintaining a clean and organized workspace
- KU7.** safety regulations and use of PPE during maintenance activities

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain clear and concise maintenance records
- GS2.** communicate maintenance requirements to supervisors or maintenance teams
- GS3.** follow checklists and schedules for systematic upkeep
- GS4.** apply problem-solving skills to troubleshoot basic machine issues
- GS5.** adhere to workplace safety and environmental policies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Perform routine maintenance of CNC milling machine</i>	15	22	-	-
PC1. clean the machine bed, fixtures, and tool holders after operations	3	4	-	-
PC2. check and top up coolant and lubrication levels as per maintenance schedule	3	5	-	-
PC3. inspect belts, filters, and hoses for wear and tear	3	4	-	-
PC4. verify spindle alignment and machine accuracy using test indicators	3	4	-	-
PC5. lubricate moving parts following manufacturer's recommendations	3	5	-	-
<i>Maintain cutting tools and accessories</i>	12	18	-	-
PC6. inspect cutting tools for wear, chipping, or deformation	3	4	-	-
PC7. replace worn-out tools and update tool inventory records	3	5	-	-
PC8. store tools in a clean, dry, and organized manner to prevent damage	3	4	-	-
PC9. calibrate and store measuring instruments securely	3	5	-	-
<i>Maintain cleanliness and safety of the work area</i>	13	20	-	-
PC10. remove chips, dust, and debris from the work area regularly	3	4	-	-
PC11. segregate and dispose of plastic waste and scrap as per guidelines	3	4	-	-
PC12. ensure all safety guards, covers, and emergency stops are functional	3	4	-	-
PC13. store hazardous materials in compliance with safety norms	2	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. maintain proper lighting and ventilation in the work area	2	4	-	-
NOS Total	40	60	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	RSC/N9903
NOS Name	Maintain machine, tools, and work area
Sector	Rubber
Sub-Sector	Plastic Post - Processing
Occupation	CNC Machining and Die Manufacturing
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	30/04/2025
Next Review Date	30/04/2028
NSQC Clearance Date	30/04/2025

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RSC/N5007: Follow health & safety protocol

Description

This unit is about maintaining health and safety of self and others at workplace.

Scope

The scope covers the following :

- Maintain a clean and efficient workplace
- Render appropriate emergency procedures
- Maintain standard safety procedures at the workplace
- Participate in safety awareness campaigns
- Understand potential sources of accidents
- Use safety gears to avoid accidents

Elements and Performance Criteria

Maintain a clean and efficient workplace

To be competent, the user/individual on the job must be able to:

- PC1.** undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor
- PC2.** work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy
- PC3.** read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc
- PC4.** prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.
- PC5.** use equipment and materials safely and correctly and return the same to designated storage when not in use
- PC6.** dispose off waste safely and correctly in a designated area
- PC7.** risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace
- PC8.** perform work in a manner which minimizes environmental damage
- PC9.** all procedures and work instructions for controlling risk are followed closely
- PC10.** report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger

Render appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

- PC11.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.
- PC12.** follow emergency procedures as per company standards and workplace requirements

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- PC13.** use emergency equipment in accordance with manufacturers' specifications and workplace requirements
- PC14.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.
- PC15.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC16.** dispose off medical waste in accordance with workplace requirements
- PC17.** report details of first aid administered in accordance with work place procedures

Maintain standard safety procedures at the workplace

To be competent, the user/individual on the job must be able to:

- PC18.** comply with general safety procedures
- PC19.** follow standard safety procedures while handling equipment, hazardous material or tool
- PC20.** check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.
- PC21.** ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure
- PC22.** keep the workplace organized, swept, clean and hazard free

Participate in safety awareness campaigns

To be competent, the user/individual on the job must be able to:

- PC23.** attend fire drills and other safety related workshops organized at the workplace
- PC24.** be aware of first aid, evacuation and emergency procedures
- PC25.** be alert of any events and do not be negligent to any safety procedures to be followed

Understand potential sources of accidents

To be competent, the user/individual on the job must be able to:

- PC26.** avoid accidents while using hazardous chemicals, machines, sharp tools and equipment

Use safety gears to avoid accidents

To be competent, the user/individual on the job must be able to:

- PC27.** use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)
- PC28.** handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** policies on incentives, delivery standards, and personnel management.
- KU2.** occupational safety and health policy followed
- KU3.** emergency evacuation procedure
- KU4.** medical policy
- KU5.** company laws and acts
- KU6.** the risks to health and safety and the measures to be taken to control those risks in the area of work

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- KU7.** workplace procedures and requirements for the handling of workplace injuries/illnesses.
- KU8.** basic emergency first aid procedure
- KU9.** local emergency services
- KU10.** reporting on accidents, incidents and problems to appropriate authorities.
- KU11.** how to use machines as per standard operating procedure
- KU12.** how to maintain work area safe and secure
- KU13.** use of hazardous materials, tools and equipments
- KU14.** emergency evacuation and first aid procedures to be followed
- KU15.** personal hygiene and fitness requirements
- KU16.** general duties under the relevant health and safety legislation
- KU17.** what personal protective equipment and clothing should be worn and how it is cared for
- KU18.** the correct and safe way to use materials and equipment required for work
- KU19.** the importance of good housekeeping in the workplace
- KU20.** safe disposal methods for waste
- KU21.** methods for minimizing environmental damage during work

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record data which are required for record keeping purpose
- GS2.** report problems to the appropriate person in a timely manner
- GS3.** write descriptions and details about incidents in reports
- GS4.** read instruction manuals for hand tools and equipment
- GS5.** read instructions on work orders and procedures
- GS6.** receive instructions and seek advice from superiors
- GS7.** communicate clearly and effectively with others
- GS8.** take a decision for any change/issue based on earlier successes(documented previous history)on similar issues
- GS9.** work out changes in case a new improved machine/equipment is added in the process or any new material/chemical is developed replacing existing one.
- GS10.** make changes in cycle time due to improved process.
- GS11.** use the standard operating procedure or trouble shooting manuals for trouble shooting and other reference documents approved by plant management
- GS12.** consult the peer group and superiors to arrive at a favourable decision.
- GS13.** use of standard available problem solving techniques for decision making
- GS14.** review and analyze the process steps to check on system non adherence andnon conformity
- GS15.** review the current sop and other standards for continuous improvement to facilitate decision making
- GS16.** take a calculated risk with minimum losses
- GS17.** schedule daily activities and drawing up priorities; allocate start times, estimation of completion times and materials, equipment and assistance required for completion.

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- GS18.** match customer needs/specification by adjusting the processing conditions (interact with customer in case any clarification required)
- GS19.** ensure that performance of his action/operation/activity does not lead to any divergence from the specified quality of the final product as required by the customer.
- GS20.** complete the assigned task in timely manner so that the final product is delivered in the timeline given by the customer.
- GS21.** communicate effectively to the superior/customer for any delay in supplies to the clients.
- GS22.** work towards fulfilling the customers requirement as per their demand.
- GS23.** in case of any complaint, ensure its timely resolution if the problem is emanating at his level
- GS24.** communicate effectively to the superior/customer for any delay in resolving the problem faced by the customer.
- GS25.** maintain good/cordial relation with customers.
- GS26.** work on the feedback received from customer regarding the product.
- GS27.** use first aid treatment in case of any injury/accident.
- GS28.** monitor and maintain the condition of tools and equipment
- GS29.** assess situation & identify appropriate control measures
- GS30.** act, communicate and report in emergency situation

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain a clean and efficient workplace</i>	9	21	-	-
PC1. undertake basic safety checks before operation of all machinery and equipment and report hazards to the appropriate supervisor	2	2	-	-
PC2. work for which protective clothing or equipment is required is identified and the appropriate protective clothing or equipment is used in performing these duties in accordance with workplace policy	2	2	-	-
PC3. read and understand the hazards of use and contamination mentioned on the labels of chemicals, utilities etc	-	2	-	-
PC4. prior to performing manual handling jobs, risk is assessed and work is carried out according to currently recommended safe practices.	2	2	-	-
PC5. use equipment and materials safely and correctly and return the same to designated storage when not in use	1	2	-	-
PC6. dispose off waste safely and correctly in a designated area	2	4	-	-
PC7. risks to bystanders are recognized and action taken to reduce risk associated with jobs in the workplace	-	2	-	-
PC8. perform work in a manner which minimizes environmental damage	-	2	-	-
PC9. all procedures and work instructions for controlling risk are followed closely	-	1	-	-
PC10. report any accidents, incidents or problems without delay to an appropriate person and take immediate necessary action to reduce further danger	-	2	-	-
<i>Render appropriate emergency procedures</i>	9	18	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to emergency.	2	4	-	-
PC12. follow emergency procedures as per company standards and workplace requirements	2	4	-	-
PC13. use emergency equipment in accordance with manufacturers' specifications and workplace requirements	2	4	-	-
PC14. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques.	-	1	-	-
PC15. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	2	-	-
PC16. dispose off medical waste in accordance with workplace requirements	-	1	-	-
PC17. report details of first aid administered in accordance with work place procedures	3	2	-	-
<i>Maintain standard safety procedures at the workplace</i>	6	15	-	-
PC18. comply with general safety procedures	2	4	-	-
PC19. follow standard safety procedures while handling equipment, hazardous material or tool	-	2	-	-
PC20. check parts of the workplace and take preventive actions like spraying and other steps to protect from leakages, water logging, pests, fire, pollution, etc.	2	4	-	-
PC21. ensure no accidents and damages at the workplace, reporting of any breach of company safety procedure	-	1	-	-
PC22. keep the workplace organized, swept, clean and hazard free	2	4	-	-
<i>Participate in safety awareness campaigns</i>	2	8	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. attend fire drills and other safety related workshops organized at the workplace	1	3	-	-
PC24. be aware of first aid, evacuation and emergency procedures	1	3	-	-
PC25. be alert of any events and do not be negligent to any safety procedures to be followed	-	2	-	-
<i>Understand potential sources of accidents</i>	1	3	-	-
PC26. avoid accidents while using hazardous chemicals, machines, sharp tools and equipment	1	3	-	-
<i>Use safety gears to avoid accidents</i>	3	5	-	-
PC27. use safety materials such as protective gear, goggles, caps, shoes, etc.(as applicable with workplace)	2	2	-	-
PC28. handle heavy and hazardous materials with care and using appropriate tools and handling equipment such as trolleys, ladders	1	3	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	RSC/N5007
NOS Name	Follow health & safety protocol
Sector	Rubber
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2026
Next Review Date	18/02/2029
NSQC Clearance Date	18/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
RSC/N9901.Interpret CNC programs and drawings and set up CNC Milling Machine	40	60	-	-	100	35
RSC/N9902.Perform milling operations on plastic workpieces using CNC Machine	40	60	-	-	100	40
RSC/N9903.Maintain machine, tools, and work area	40	60	-	-	100	15
RSC/N5007.Follow health & safety protocol	30	70	-	-	100	5
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	170	280	-	-	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.